

Generic Feedback on the January Assessment for 2LA/2LALP

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General comments

Overall the class again did very well, the mean mark is towards the higher end of the expected interval. The quality of explanation, which is what I was looking for in the solutions, was mostly adequate. Of course, in computational problems many students, as usual, reduced their solutions to a sequence of formulas, without much comment about what is being done. This is not so good and may lead to a loss of marks, for example, due to misunderstanding. A proper solution should be self-contained, and the reader shouldn't be guessing what is being done in each problem.

The exam included definitions and standard proofs. In definitions, I was looking for a detailed text specifying all the ingredients, not just listing the axioms. In proofs, I was looking for the overall logic, and that no case is missed.

General comments about the questions

Question 1: (a)(i) is a standard definition; state what U and V are and mention the field of scalars before listing the axioms of linearity; (a)(2) can be done either by definition or by one step check and both were used; those who wrote the polynomials in full typically ended up with a longer argument; those who simply used the properties of differentiation had shorter proofs. Students had a bit of a problem with part (b), but it did help that the matrix was given explicitly. Part (c)(i) was a standard definition and comments are generally the same as for (a)(i): specify all ingredients of the definition for full marks. Part (c)(ii) was done well, especially by those who used the determinant check. Part (d)(i) is again a standard definition and (d)(ii) is a standard proof. Not all students attempted it, but those who did mostly did it well; part (d)(iii) was to check the overall understanding of the topic, and some students could not see the relevance of part (ii) and the computation in (b). Part (e) was entirely computational and mostly done well. I am happy to see that most students understood the point about the eigenspace being a subspace, and so listing eigenvectors using the span notation.

Question 3: Part (a) included two standard definitions, so the usual criteria apply. Part (b) was checking students understanding of what the Gram matrix is, and it was mostly done well. Part (c) included a standard definition and a standard proof from the later part of the course, and quite a few students did not seem to study those parts as well as they did the earlier parts. Equally, in (c), quite a few students found it difficult to apply the geometric concepts of length, angle and projection in strange inner product spaces, such as the space of polynomials. So this is something that I will spend time more next year. However, in (c), where student had to compute in terms of the 4-dimensional version of the dot product, they were a lot more comfortable and mostly applied Gram-Schmidt process well.

Where students typically did well

In all computational problems, especially where the problem was formulated in terms of matrices or familiar column/row spaces and dot product. Definitions were done quite well, too.

Typical areas for improvement

Most importantly, write more explanation, not just the calculation. I think quality of the texts has improved this year, so we can now strive for a higher standard. Do more examples involving “strange” vector spaces and inner product spaces, especially from the final chapter.